This CVPR Workshop paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

Fairness-Aware Boosting model for Imbalanced 3D Point Cloud Segmentation in Autonomous Driving

Elahe Yahyapour
University of Massachusetts, Amherst
Amherst, MA, USA
eyahyapourde@umass.edu

Chengbo Ai
University of Massachusetts, Amherst
Amherst, MA, USA
chengbo.ai@umass.edu

Abstract

Large-scale 3D point clouds are essential for autonomous driving but suffer from a significant class imbalance, where certain object categories dominate while others are underrepresented. This imbalance often causes the model to be biased toward classes with the most data or distinct physical shapes. Consequently, when majority classes surround underrepresented classes, the model becomes confused (e.g., road markings adjacent to the ground) and performs poorly. To address this fairness issue, we propose a new approach that is an enhanced version of RandLA-Net using an ensemble method, AdaBoost, which adaptively reviews misclassified samples to improve performance for underrepresented classes overshadowed by majority classes. then, we employed a fairness-aware evaluation which is weighted Intersection over Union (wIoU) metric to emphasize underrepresented classes, ensuring a fair evaluation of segmentation performance across all classes. Experimental results demonstrate that the RandLA-NetBoosting outperforms the original RandLA-Net, mitigating class imbalance bias and improving overall segmentation accuracy. This illustrates the potential of fairness-aware methods to address real-world challenges in autonomous driving.

1. Introduction

Deep learning models play a crucial role in the semantic segmentation of 3D point clouds by identifying and differentiating various objects within a region of interest, whether in robotics, autonomous driving [2, 4, 5], urban analysis, or remote sensing [18, 20]. In the domain of reliable self-driving vehicles, the model must accurately identify both frequent and infrequent objects (e.g., vehicles, road markings, pedestrians, road signs) to make accurate real-time decisions. However, many real-world datasets contain classes with a high frequency of labeled points, such as ground or building, while rare but safety-critical classes, such as road marking or fence, are underrepresented classes that share similar physical attributes to their surrounding classes, causing models to become biased toward the more frequent classes and those with distinct physical attributes, rendering them unable to detect the minority classes. These misdetections compromise the safety of vulnerable road users and contribute to accidents in autonomous driving [2, 4, 5].

The correlation between segmentation model performance and crashes has previously been discussed in simulation-based and field studies [4, 5]. Perception systems driven primarily by segmentation models have been forming the backbone of modern autonomous driving pipelines. These models are responsible for obstacle detection, lane marking recognition, and understanding other objects/agents in the environment. When underrepresented classes with similar physical attributes to the majority classes are not accurately segmented, the autonomous driver's planning and control mechanism may face a lack of crucial information, which results in improper maneuvers or failures to avoid collisions.

Although missegmentations may rarely be the sole cause of crashes, recent high-profile incidents indicate that perception failures can be a significant contributing factor [4, 5]. Simulation systems like CARLA [5] have indicated that improving the performance of segmentation models, especially for rare classes, will increase safer driving behaviors. Field data from industrial and commercial fleets have shown certain types of collisions that have a strong correlation with misclassifications in the perception pipeline [2, 4]. These findings emphasize the importance of semantic segmentation models that can effectively address misclassified underrepresented classes with attributes similar to majority classes in the real world.

In 3D point clouds, segmentation becomes even more challenging due to the spatial distribution and the large number of points involved. Early works like PointNet

[13] and its hierarchical extension PointNet++ [14] pioneered learning unordered 3D point clouds, but these methods suffer from limitations in scalability when handling massive amounts of data in a complex environment. RandLA-Net[10] developed an efficient and powerful method to handle the scalability problem with a random sampling strategy that preserves essential geometric information while enabling near-real-time inference. Using the Local Feature Aggregation (LFA) module, RandLA-Net effectively encodes both local and global context without exhausting neighbor searches.

Although RandLA-Net outperforms other models such as DGCNN (Dynamic Graph CNN) [17], KPConv (Kernel Point Convolution) [16], Sparse Convolutional Networks (SSCNs) [7], or PointTransformer [19], in large-scale segmentation tasks, it still struggles with imbalanced datasets. This becomes particularly problematic in autonomous driving, where errors in detecting underrepresented classes, such as lane marking, can lead to significant risks.

In this paper, we propose enhancing RandLA-Net with an ensemble method, including ADABoosting [6] to re-weight the misclassified samples to emphasize those from minority classes. Then we replace the traditional mean IoU with a fairness-aware evaluation using a weighted Intersection over Union (wIoU) metric [3], ensuring that infrequent classes strongly affect both training and final performance measurement. Together, these strategies produce a RandLA-Net variant, that is, RandLA-NetBoosting, that maintains its efficiency while improving the segmentation accuracy for underrepresented categories.

The following parts of this paper include Section 2, a review of the literature on point-cloud segmentation and existing solutions to imbalanced classes. Section 3 introduces the data set that we used and Section 4 demonstrates details of our proposed method, which is boosting-based training and weighted IoU. Sections 5 and 6 present the fairness awareness metric and experimental results and state-of-the-art comparisons, and Sections 7 and 8 include discussion and conclude with directions for future work.

2. Literature review

The 3D point cloud segmentation models have evolved significantly in the past decade, with early models such as PointNet [13] and its successor, PointNet++ [14], which focus on directly processing raw point-cloud data. PointNet is designed to handle unordered points; however, it faces limitations in capturing fine-grained local detail. Then, PointNet++ improves performance by introducing hierarchical feature extraction but still struggles with high computational cost. Both methods do not address class imbalance. Other methods such as DGCNN [17], KPConv (Kernel Point Convolution) [16], Sparse Convolutional Networks (SSCNs) [7], or PointTransformer [19], focus on improving local feature learning and achieving computational efficiency, but face challenges with large or dense datasets, none of them inherently tackles the class imbalance problem.

Among these methods, RandLA-Net [10], introduced by Hu et al., stands out for its capacity to process large-scale 3D point clouds in real time while preserving essential spatial information. RandLA-Net uses random sampling to minimize computational load and employs a local feature aggregation module, making it suitable for large autonomous driving datasets. However, the aggregation of feature based on attention in the network can underestimate direct geometric relationships, causing suboptimal feature fusion in dense or complex environments. Additionally, it does not inherently resolve class imbalance, leaving the model vulnerable to skewed segmentation outcomes when minority classes are significantly underrepresented and surrounded by majority classes with similar physical attributes. This can lead to biased predictions and safety concerns in applications such as autonomous driving.

Ensemble methods such as AdaBoost focus on overcoming data imbalance through boosting algorithms. They have shown promising results in addressing imbalanced data by adaptively re-weighting misclassified samples, directing the learning process toward those difficult instances. AdaBoost, introduced by Freund and Schapire [6], has been successful in various classification tasks by focusing on hard-to-classify training instances. By re-weighting misclassified samples, AdaBoost helps the model concentrate on the minority classes that are often overshadowed by majority classes or those that have a distinct physical shape, making it more sensitive to subtle distinctions between classes. This characteristic is particularly useful in 3D point cloud segmentation, making it suitable to integrate with RandLA-Net to enhance segmentation accuracy and fairness in autonomous driving.

Recent studies highlight the challenges of class imbalance in 3D segmentation, such as those of Pan et al. [12] and Han et al. [9], which underline the difficulties models face due to skewed class distribution and the necessity of adaptive methods. Although some propose solutions like subspace prototype guidance or semi-supervised learning, these methods still face challenges with intra-class variability and computational cost. Barnefske and Sternberg [1] emphasize how imbalanced datasets impair minority class recognition in PointNet, endorsing adaptive methods like re-weighting or ensemble techniques to achieve more equitable segmentation. Li et al. [11] introduce a semi-supervised

method to address labeling scarcity and class imbalance, but their model requires careful parameter tuning of both supervised and unsupervised objectives. By combining the ensemble method, AdaBoosting, with RandLA-Net, our approach directly tackles these issues—especially fairness in class imbalanced datasets—and ensures more fair segmentation and better performance on imbalanced data, providing a promising pathway for real-world 3D perception tasks in autonomous driving.

3. Dataset

To evaluate our model, we used two complementary datasets, Toronto 3D [15] and Semantic3D [8]. These datasets together provide a comprehensive assessment of 3D point cloud segmentation in diverse real-world scenarios, from dense urban roadways to expansive outdoor terrains.

Toronto 3D is an important resource for understanding urban scenes, covering one kilometer (i.e., 0.62 mile) of city roads with 78.3 million points. The density of this dataset is moderate and provides more than 8000 annotated frames with eight object classes including cars, road markings, buildings, and fences, as well as an unclassified class. Their LiDAR data capture all measured points, including those beyond a fixed distance, and offer richer scene information. Despite its urban focus, the medium size of Toronto 3D and potential LiDAR-related challenges, such as occlusions, make it suitable for evaluating model performance in complex autonomous driving environments.

Semantic3D is a large-scale benchmark that offers over 4 billion points captured from terrestrial scanners in diverse outdoor settings with categories such as cars, buildings, and natural features. It is perfect for assessing the capacity of the models to handle large data volumes and complex environmental factors such as angle variation and occlusion. The predefined training and testing split of the data set supports the reliability and facilitates comparisons, while its crowded scenes expose challenges in real world environments such as cluttered objects and variation in point density. These characteristics make Semantic3D particularly valuable for assessing the robustness of segmentation models in large-scale outdoor settings.

The distribution of these two datasets is shown in figure 1, which illustrates how certain classes, such as Ground and Building in Toronto 3D, and building and ma-made Terrian in Semantic3D, dominate over other classes like Road Marking and Fence in Toronto 3D and Cars and Scanning Artifacts in Semantic3D.

By examining our model on these two datasets, our experiment evaluates RandLA-NetBoosting's performance across a range of point cloud densities and scene complexities. Evaluating these dual datasets helps to ensure the generalizability and robustness of our fairness-enhanced model under realistic autonomous driving conditions.

4. Methodology

4.1. RandLA-Net Backbone

Our proposed methods, RandLA-NetBoosting, build on the effective point cloud segmentation architecture of RandLA-Net by integrating ensemble boosting techniques, AdaBoost, to address the challenges of class imbalance and fairness in autonomous driving datasets. In our work, the RandLA-Net is first used to extract robust local and global features from the raw point cloud data. Then these features are fed to an ensemble method of boosting classifiers that iteratively re-weight training sample points to explicitly address fairness challenges caused by class imbalance in large-scale 3D point clouds for autonomous driving. This imbalance is specifically crucial for classes that share similar physical attributes with adjacent, more frequent classes (e.g., road marking vs. surrounding ground), where subtle distinctions are easily overlooked. An overview of the architecture of our proposed model is provided in Figures 2 .

The RandLA-Net backbone is used to extract robust point-wise features from a raw 3D point cloud, which addresses the challenges of handling sparse and irregular data with high computational efficiency.

We denote the raw 3D point cloud as a set of (N) points:

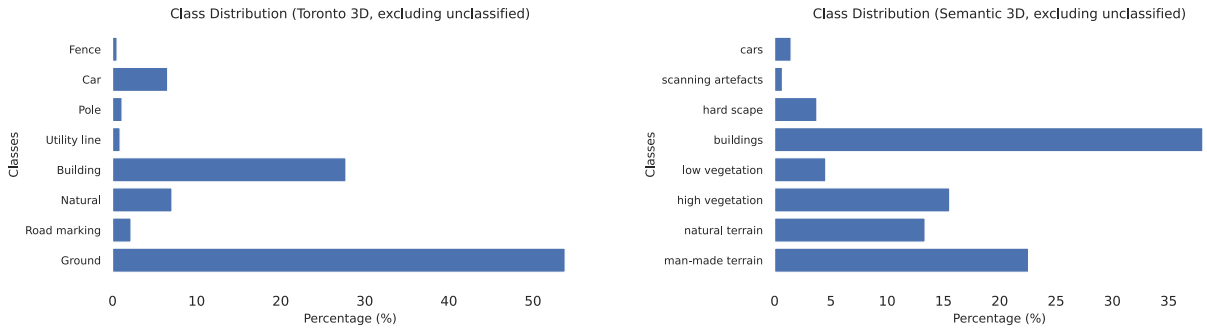
$$\mathcal{P} = \mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N \tag{1}$$

Each point $\mathbf{x}_i$ includes its 3D spatial coordinates($\mathbf{x}_i \in \mathbb{R}^3$) and may also consist of additional attributes, such as color, intensity, or return number, that depend on the sensor capabilities. The RandLA-Net framework is based on an encoder-decoder that implements random sampling to reduce the computational burden of processing large-scale point clouds while preserving important spatial information. This architecture is combined with dilated residual blocks, designed to capture both local and global context without requiring an exhaustive neighborhood search.

The encoder part of RandLA-Net builds a local neighborhood for each point in the point cloud by computing a Relative Positional Encoding (RPE) that captures the spatial relationship between points:

$$\mathbf{r}_{ij} = \mathbf{x}_i - \mathbf{x}_j, \quad d_{ij} = |\mathbf{r}_{ij}|_2 \tag{2}$$

In here, $\mathbf{r}_{ij}$, is a vector that demonstrates the difference between $\mathbf{x}_i$, $\mathbf{x}_j$, and d_{ij} in the Euclidean distance



(a) Toronto 3D

(b) Semantic3D

Figure 1. Distribution of the Toronto 3D and Semantic3D datasets per class (excluding unclassified) in percentage.

between them. The Relative positional encoding (RPE) $(\mathbf{x}_i, \mathbf{x}_j)$ is calculated as below:

$$\text{RPE}(\mathbf{x}_i, \mathbf{x}_j) = [d_{ij}, \mathbf{r}_{ij}, \mathbf{x}_i, \mathbf{x}_j] \tag{3}$$

This encoding captures both the distance between points and their relative positions and the aggregated via an attention-based pooling operation to yield local rich features that are invariant to the order of points.

The decoder part of RandLA-Net processes these local features and progressively down-samples the point cloud through multiple layers, each incorporating a residual connection. After several downsampling steps, the decoder reconstructs the per-point features, resulting in the final logits:

$$\mathbf{Z} \in \mathbb{R}^{N \times C} \tag{4}$$

where (C) is the number of classes and a softmax is applied to obtain pointwise class probabilities.

4.2. Ensemble Enhancements

4.2.1 RandLA-NetBoosting

To address the class imbalance and fairness challenges that appear in autonomous driving and real-world dataset, we integrated an ensemble boosting method on top of the RandLA-Net backbone. AdaBoost is employed to re-weight the misclassified sample during training to ensure that underrepresented classes, which share similar physical attributes with adjacent majority classes, are adequately learned and not overlooked.

In the context of autonomous driving, certain critical classes, such as road marking, fences, are often underrepresented in training data and are surrounded by more frequent classes such as ground and buildings that have similar visual characteristics. This class imbalance, coupled with the similarity between classes, can cause model confusion, where the model fails to distinguish between them, resulting in poor segmentation performance. The AdaBoost algorithm addresses this issue by focusing more on misclassified points, especially those from rare classes, during each boosting iteration.

The RandLA-Net logit is $\mathbf{Z}_b \in \mathbb{R}^{N_b \times C}$, which is for a batch of N_b points covering C classes. After applying softmax to $\mathbf{Z}_b$, one achieves the predicted probabilities $\hat{\mathbf{P}}_b \in [0,1]^{N_b \times C}$. Afterward, AdaBoost treats each row of $\mathbf{Z}_b$ (or comparablely $\hat{\mathbf{P}}b$) as an input feature vector for a set of weak learners $\{h_t\}_{t=1}^{T}$. Each weak learner of $\{h_t\}_{t=1}^{T}$, in our implementation, is a shallow decision tree with limited depth (i.e., depth = 3), which ensures fast training on point-wise logits. These weak learners are used to classifying the points based on their predicted probabilities.

AdaBoost maintains a weight distribution $D_t(i)$ on all training points in iteration t. Initially, all the points are given equal weight:

$$D_1(i) = \frac{1}{N_b} \tag{5}$$

Therefore, each point i in the batch is equally important, and AdaBoost minimizes the weighted error when fitting the weak learner $(t) - th$.

$$\epsilon_t = \sum_{i=1}^{N_b} D_t(i), \mathbb{I}\big[h_t(\hat{\mathbf{p}}_i) \neq y_i\big] \tag{6}$$

where $\hat{\mathbf{p}}_i$ is the logit or probability-based feature vector of the $(i) - th$ point, and y_i is the ground truth label. The learner's contribution α_t is

$$\alpha_t = \frac{1}{2} \ln\left(\frac{1 - \epsilon_t}{\epsilon_t}\right) \tag{7}$$

The misclassified points receive increased emphasis on the next iteration by updating as follows

$$D_{t+1}(i) = \frac{D_t(i) \exp\big[-\alpha_t, \mathbb{I}\big(h_t(\hat{\mathbf{p}}_i) = y_i\big)\big]}{Z_t} \tag{8}$$

where Z_t is a normalization factor to ensure $\sum_i D_{t+1}(i) = 1$. Therefore, samples that remain misclassified by h_t have their weights add to D_{t+1}.

After T iterations, AdaBoost combines weak learners into a strong classifier.

$$H(\hat{\mathbf{p}}) = \arg\max_{k \in 1 \ldots, C} \sum_{t=1}^{T} \alpha_t \mathbb{I}\big[h_t(\hat{\mathbf{p}}) = k\big]. \quad (9)$$

During this process, RandLA-Net first produces logits or probabilities for each point, which are fed into each h_t. The final labeling is determined by a weighted majority vote among the weak learners. As AdaBoost iteratively shifts emphasis on hard-to-classify points, it is specifically adept at handling infrequent categories adjacent to frequent categories with similar physical attributes.

At test time, the logit vector $\hat{\mathbf{p}}i$ for point (i) passes through all decision trees $\{h_t\}_{t=1}^{T}$, each of them weighted by α_t. AdaBoost updates weights online during training rather than waiting until the epoch ends, allowing for a more immediate adaptation to misclassified samples. Additionally, direct usage of RandLA-Net's logits avoids the overhead of constructing strong explicit deep features or histograms, which leads to a tighter collaboration between the convolutional architecture and the boosting procedure. This design effectively provides a strong improvement for segmentation fairness and overall performance for critical but underrepresented classes.

Figure 2 provides an overview of the architecture for RandLA-NetBoostig. The overall workflow involves using RandLA-Net to extract point-wise features, followed by applying AdaBoost to enhance the performance of underrepresented classes. The combination of efficient point cloud processing and adaptive re-weighting creates a robust solution for segmentation of large-scale 3D point clouds in autonomous drives scenarios, where fairness and safety are non-negotiable.

5. Metrics

Although region-based metrics can be aggregated across all classes to calculate overall performance, they may not reflect segmentation quality in highly imbalanced scenarios, since aggregated scores can be dominated by classes with large point counts.

One of the most popular metrics widely used in semantic segmentation for large-scale point clouds is Intersection Over Union (IoU), which involves assigning each point to one of the C classes. The IoU for class c is defined as

$$\mathrm{IoU}_c = \frac{|A_c \cap B_c|}{|A_c \cup B_c|}, \quad (10)$$

Where A_c is the set of predicted points as class c, and B_c is the set of points with the grand-truth label of c. The overall segmentation performance is often summarized as the mean IoU (mIoU), which is computed as the average of IoU_c in all classes. However, in a class imbalanced dataset, which is common in autonomous driving where, for example, classes like Road or Building dominate over minority classes like road marking or fence, this average can mask poor performance on rare classes.

To address this fairness problem, we adopt a weighted IoU metric that scales the contribution of each class according to its inverse frequency. Specifically, for each class c, we calculate the weight :

$$w_c = \frac{1}{\mathrm{frequency}_c + \epsilon} \quad (11)$$

Where $\mathrm{frequency}_c$ is the number of points belonging to class c in the training set and ϵ is a small constant to avoid division by zero. These weights are then normalized to sum to one.

$$\tilde{w}c = \frac{w_c}{\sum k = 1^C w_k}, \quad (12)$$

The weighted IoU is computed as

$$\text{Weighted IoU} = \sum_{c=1}^{C} \tilde{w}_c, \mathrm{IoU}_c \quad (13)$$

Although mean IoU treats all classes equally and can cause falsely high performance when majority classes dominate the data set, on the contrary, weighted IoU indicates the impact of minority classes, which ensures that the model's imperfections in underrepresented categories are taken into account, often surrounded by overrepresented ones with similar physical attributes (such as subtle road marking adjacent to similar surfaces that are more clearly reflected). As shown in recent work by [3], this formulation allows for an evaluation that considers both region and boundary accuracy by adjusting for class frequency and boundary importance. This approach effectively promotes balanced treatment by penalizing both false positives and false negatives in minority classes and avoids overemphasizing extremely rare classes.

In our framework, this metric is aligned with the adaptive re-weighting performance of our ensemble methods, which ensures that improvements on challenging, infrequent classes lead to a solid enhancement in overall segmentation performance, which is particularly crucial in autonomous driving, where even rare misclassifications can have severe safety implications.

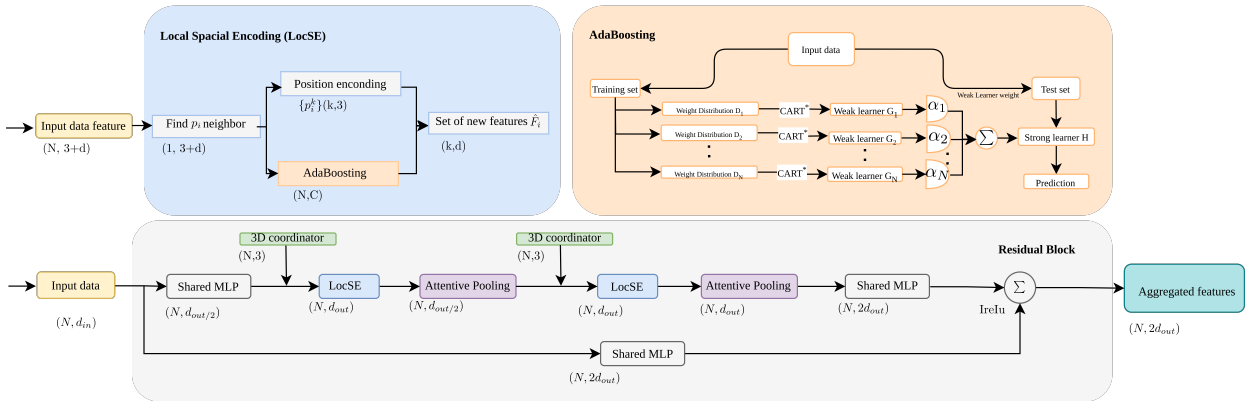


Figure 2. Enhanced RandLA-Net architecture with AdaBoosting. In our model, the AdaBoost ensemble is integrated into the LocSE block, where adaptive re-weighting is performed at each training epoch to focus on hard-to-classify samples. The network produces point-wise logits of dimension C, where C denotes the number of classes.

6. Results

In the semantic segmentation of large-scale urban scenes, class imbalance is a common issue. Figure 1 demonstrates the class distribution for the Toronto 3D dataset, where classes such as Ground dominate over 50% of the points while minority classes such as Road Marking or Fence are represented by only a few percent. This skew causes standard models like RandLA-Net to become biased toward the frequent classes, which results in poor performance for rare but crucial classes that share similar attributes with majority classes (e.g., subtle road markings adjacent to the road surface).

To achieve a better view of segmentation performance under these conditions, we adopt both Mean IoU and Weighted IoU (wIoU) as evaluation metrics. Weighted IoU is particularly useful as it emphasizes the impact of underrepresented classes, thus it reveals a better understanding of performance on rare classes that share similar physical attributes with dominant classes, which is vital for addressing fairness in class imbalanced datasets and ensuring safe autonomous driving.

To perform a fair comparison, we use the same hyperparameters for all models, and they run on an Ubuntu 22.04.5 LTS with 13th Gen Intel® Core™ i9-13900K and NVIDIA GeForce RTX 4090 GPU with CUDA V11.5.119.

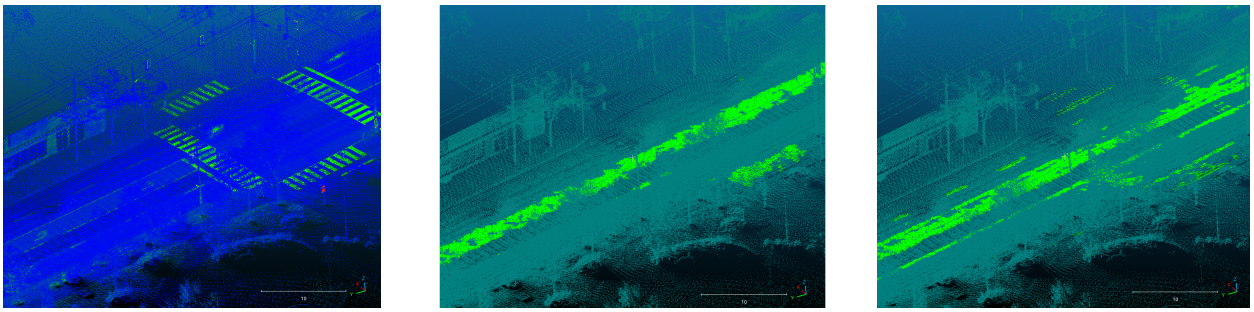
Table 1 reports the overall quantitative results on the Toronto 3D dataset based on weighted IoU. As shown in 1, our method outperforms the baseline RandLA-Net from 57.49% to 59.05%). The improvements are especially improving mean IoU for classes such as *Road Markings* (an increase from 10.74% to 13.94%) and *Fence* (from 38.13% to 43.20%), which are crucial for fairness in safety critical applications in autonomous driving.

In addition to Toronto 3D, a similar pattern is observed in the Semantic3D dataset. Figure 1 in figure b also shows the class imbalance in this dataset.

In the second Table 2, the results shown using Weighted IoU for overall performance further confirm these observations. Here, RandLA-NetBoosting achieves a Weighted IoU of 67.07%, compared to 64.77% for the baseline. The improvements are especially improving mean IoU for classes such as *Scanning Artifacts* (an increase from 60.18% to 63.99%) and *Low Vegetation* (from 33.49% to 35.42%). This clearly indicates that the AdaBoost integration better addresses the class distribution imbalance by elevating the performance in minority classes.

The visualized performance results of the baseline RandLA-Net and our model, RandLA-NetBoosting, along with the original view of the same scene from the Toronto 3D dataset, are shown in Figure 3. As seen, our proposed method, RandLA-NetBoosting, captures more details of the Road Marking in the scene, highlighting its ability to improve the performance of the baseline RandLA-Net for hard-to-classify classes.

Overall, our experiments demonstrate that our model, RandLA-NetBoosting, outperforms the original RandLA-Net by improving segmentation performance on rare classes that are adjacent to dominant classes sharing similar physical attributes. In particular, the RandLA-NetBoosting approach demonstrates significant improvements in minority classes such as Road Marking and Fence, which is essential for enhancing fairness in safety-critical autonomous driving applications.



(a) Original view of Toronto 3D (b) View by applying RandLA-Net (c) View by applying RandLA-NetBoosting

Figure 3. Three views of the same scene on Toronto 3D dataset: (a) Original scene, (b) after applying RandLA-Net, and (c) after applying RandLA-NetBoosting. As it is shown, RandLA-NetBoosting captures more details of the Road Marking in the scene.

Class	RandLANet	RandLa-NetBoosting
Ground	**87.11**	83.89
Road Markings	10.74	**13.94**
Natural	94.08	**94.86**
Building	88.11	**88.80**
Utility Line	**84.43**	83.70
Pole	**76.99**	74.12
Car	**77.56**	76.18
Fence	38.13	**43.20**
Weighted IoU	57.49	**59.05**

Table 1. Quantitative results on the Toronto 3D dataset based on Weighted IoU. In each column, the highest value is highlighted in bold.

Class	RandLa-Net	RandLa-NetBoosting
Man-made Terrain	95.65	**95.69**
Natural	**85.04**	82.81
High Vegetation	**87.04**	84.64
Low Vegetation	33.49	**35.42**
Building	83.01	**83.43**
Hard Scape	24.69	**25.04**
Scanning Artefacts	60.18	**63.99**
Car	93.17	**94.31**
Weighted IoU	64.77	**67.07**

Table 2. Quantitative results on the Semantic3D dataset based on Weighted IoU. In each column, the highest value is highlighted in bold

7. Discussion

Our experiments show that class imbalance is a significant challenge in large-scale 3D point cloud segmentation for autonomous driving. As demonstrated by the class distribution in figure 1 majority classes such as Ground dominate the dataset, while critical minority classes (e.g., Road Marking and Fence in Toronto 3D and Scanning Artifacts and Low Vegetation in Semantic 3D) are notably underrepresented. This class imbalance biases the original RandLA-Net toward the dominant classes or classes with distinct physical attributes, resulting in poor performance on the rare classes that share similar attributes with adjacent dominant classes. For example, distinguishing between road marking and the ground becomes particularly challenging due to spatial proximity and similar appearance. Thus, the original RandLa-Net model could compromise the overall safety and reliability of autonomous driving systems where precise localization using road markings is crucial.

To address this fairness issue, our proposed RandLA-NetBoosting framework incorporates an ensemble method, AdaBoost, which introduces an additional layer of adaptability to the base model. The key advantage of ensemble methods like AdaBoost lies in their ability to focus on hard-to-classify instances during training. In our case, AdaBoost effectively reweighed misclassified instances and ensured that underrepresented classes received more attention during both the training and the enhancement process. The re-weighting mechanism enhances the learning of underepresented classes by introducing a higher penalty in both training loss via the weighted cross-entropy that improves the decision boundaries around these classes. This adjustment results in more accurate and balanced segmentation performance with addressing the biases of the original model.

One of the primary reasons for selecting AdaBoost for our framework is its ability to update sample weights in real time. This dynamic re-weighting enables AdaBoost to rapidly focus on the most challenging or hard-to-classify points. For example, features such as road marking adjacent to the ground and AdaBoost's ability to emphasize misclassified points during each iteration ensure that the model becomes more sensitive to those fine distinctions. By improving the learning of these critical areas, AdaBoost helps mitigate the bias of the model

towards dominant classes and refines its performance in rare classes.

Our quantitative results on the Toronto 3D dataset (Tables 1) and preliminary findings on the Semantic3D dataset (Table 2) consistently show that the ensemble methods significantly outperform the original RandLa-Net. RandLA-NetBoosting, using Adaboost, demonstrates substantial improvements in the segmentation of underrepresented classes. For example, road markings from 10.74% to 13.94% and for the fence from 38.13% to 43.20% in the Toronto 3D dataset and similar pattern in Sematic3D in the scanning artifact of 60.18% to 63.99% and low vegetation from 33.49% to 35.42%. These improvements are crucial to ensure fairness in autonomous driving scenarios, where accurate detection of these minority classes can have a direct impact on safety.

A key point to highlight is that our framework not only improves performance across different datasets but also specifically boosts the accuracy of classes with subtle, hard-to-classify features that are critical for real-world autonomous driving applications. Our approach excels in distinguishing between classes that share physical attributes with adjacent classes, such as road markings near the ground or fences near the building. This ability to identify and correctly segment these classes helps to create a more reliable and fair perception system for autonomous driving.

Our similar performance patterns that are observed in both the Toronto 3D and Semantic3D datasets further validate the robustness and generalizability of our ensemble-based method. By combination of efficient feature extraction of RandLA-Net with the adaptive re-weighting mechanism of AdaBoost, we directly address the fairness problem that inherent on class imbalanced datasets. This approach not only improves overall segmentation and performance, but also enhances the recognition of minority classes, to contribute to a safer and more robust autonomous driving system. Our results highlight the potential of fairness-aware methods to significantly improve the accuracy and reliability of point cloud segmentation, to ensure that all classes, especially those critical to safety, are treated fairly during training and evaluation.

Overall, our framework offers a significant step forward in addressing the fairness issues associated with class imbalance in 3D point cloud segmentation. By improving RandLa-Net with AdaBoost, we provide a powerful solution to improve both segmentation accuracy and fairness, which ultimately contributes to a safer autonomous driving application.

8. Conclusion

In this paper, we present a fairness-aware, boosting-based segmentation framework that is tailored for large-scale 3D point clouds in autonomous driving. By integrating an ensemble method, AdaBoost, with the RandLA-Net backbone, our approach explicitly addresses the pervasive class imbalance in urban datasets. This class imbalance often causes a bias toward the dominant classes, leading to poorer segmentation performance for rare classes that share similar physical attributes adjacent to dominant classes with subtle features such as road marking and fences, which are crucial for safe autonomous driving.

Our framework leverages adaptive re-weighting to bring the focus of the model to hard-to-classify samples. AdaBoost, with its online weight update, rapidly emphasizes misclassified instances during training. Furthermore, our adaptation of a weighted Intersection over Union (wIoU) metric, where each class's contribution is scaled by its inverse frequency, provides a more balanced evaluation of segmentation performance that ensures that the improvements in underrepresented classes are clearly captured.

The experimental results on the Toronto 3D and Semantic3D datasets demonstrate that our proposed method, RandLA-NetBoosting, outperforms the original RandLA-Net, which shows notable improvements in challenging classes such as Road Marking and Fence. The results of the Semantic 3D dataset reveal a similar pattern, which further validates the robustness and generalizability of our approach.

In general, our hybrid learning framework based on boosting significantly enhances segmentation fairness and overall performance in class imbalanced datasets, a critical requirement for autonomous driving applications. Further work will aim to optimize the ensemble components and extend our validation to additional real-world datasets to strengthen the safety and reliability of the perception system in autonomous vehicles.

References

[1] Eike Barnefske and Harald Sternberg. Evaluation of class distribution and class combinations on semantic segmentation of 3d point clouds with pointnet. *Ieee Access*, 11: 3826–3845, 2022. 2

[2] Chenyi Chen, Ari Seff, Alain Kornhauser, and Jianxiong Xiao. Deepdriving: Learning affordance for direct perception in autonomous driving. In *Proceedings of the IEEE international conference on computer vision*, pages 2722–2730, 2015. 1

[3] Yeong-Jun Cho. Weighted intersection over union (wiou) for evaluating image segmentation. *arXiv preprint arXiv:2107.09858*, 2021. 2, 5

[4] Felipe Codevilla, Matthias Müller, Antonio López, Vladlen Koltun, and Alexey Dosovitskiy. End-to-end driving via conditional imitation learning. In *2018 IEEE international conference on robotics and automation (ICRA)*, pages 4693–4700. IEEE, 2018. 1

[5] Alexey Dosovitskiy, German Ros, Felipe Codevilla, Antonio Lopez, and Vladlen Koltun. Carla: An open urban driving simulator. In *Conference on robot learning*, pages 1–16. PMLR, 2017. 1

[6] Yoav Freund, Robert E Schapire, et al. Experiments with a new boosting algorithm. In *icml*, pages 148–156. Citeseer, 1996. 2

[7] Benjamin Graham, Martin Engelcke, and Laurens Van Der Maaten. 3d semantic segmentation with submanifold sparse convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 9224–9232, 2018. 2

[8] Timo Hackel, N. Savinov, L. Ladicky, Jan D. Wegner, K. Schindler, and M. Pollefeys. SEMANTIC3D.NET: A new large-scale point cloud classification benchmark. In *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, pages 91–98, 2017. 3

[9] Jiawei Han, Kaiqi Liu, Wei Li, and Guangzhi Chen. Subspace prototype guidance for mitigating class imbalance in point cloud semantic segmentation. *arXiv preprint arXiv:2408.10537*, 2024. 2

[10] Qingyong Hu, Bo Yang, Linhai Xie, Stefano Rosa, Yulan Guo, Zhihua Wang, Niki Trigoni, and Andrew Markham. Randla-net: Efficient semantic segmentation of large-scale point clouds. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11108–11117, 2020. 2

[11] Mengtian Li, Shaohui Lin, Zihan Wang, Yunhang Shen, Baochang Zhang, and Lizhuang Ma. Class-imbalanced semi-supervised learning for large-scale point cloud semantic segmentation via decoupling optimization. *arXiv preprint arXiv:2401.06975*, 2024. 2

[12] Yancheng Pan, Fan Xie, and Huijing Zhao. Understanding the challenges when 3d semantic segmentation faces class imbalanced and ood data. *IEEE Transactions on Intelligent Transportation Systems*, 24(7):6955–6970, 2023. 2

[13] Charles R Qi, Hao Su, Kaichun Mo, and Leonidas J Guibas. Pointnet: Deep learning on point sets for 3d classification and segmentation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 652–660, 2017. 2

[14] Charles Ruizhongtai Qi, Li Yi, Hao Su, and Leonidas J Guibas. Pointnet++: Deep hierarchical feature learning on point sets in a metric space. *Advances in neural information processing systems*, 30, 2017. 2

[15] Weikai Tan, Nannan Qin, Lingfei Ma, Ying Li, Jing Du, Guorong Cai, Ke Yang, and Jonathan Li. Toronto-3d: A large-scale mobile lidar dataset for semantic segmentation of urban roadways. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition workshops*, pages 202–203, 2020. 3

[16] Hugues Thomas, Charles R Qi, Jean-Emmanuel Deschaud, Beatriz Marcotegui, François Goulette, and Leonidas J Guibas. Kpconv: Flexible and deformable convolution for point clouds. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 6411–6420, 2019. 2

[17] Yue Wang, Yongbin Sun, Ziwei Liu, Sanjay E Sarma, Michael M Bronstein, and Justin M Solomon. Dynamic graph cnn for learning on point clouds. *ACM Transactions on Graphics (tog)*, 38(5):1–12, 2019. 2

[18] Liangpei Zhang, Lefei Zhang, and Bo Du. Deep learning for remote sensing data: A technical tutorial on the state of the art. *IEEE Geoscience and remote sensing magazine*, 4(2):22–40, 2016. 1

[19] Hengshuang Zhao, Li Jiang, Jiaya Jia, Philip HS Torr, and Vladlen Koltun. Point transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 16259–16268, 2021. 2

[20] Xiao Xiang Zhu, Devis Tuia, Lichao Mou, Gui-Song Xia, Liangpei Zhang, Feng Xu, and Friedrich Fraundorfer. Deep learning in remote sensing: A comprehensive review and list of resources. *IEEE geoscience and remote sensing magazine*, 5(4):8–36, 2017. 1